Independent graph review

Independent retained-graph check

175 vertices − 61 components = 114 edges

E = V − C · zero cycles · 420 dispositions match

Scope: deduplicated degree-two, distinct-root edges only

Excluded labels / duplicates are not proved incapable of constraints

The root agent independently recomputed all 420 case dispositions from 60 archived known-answer records using exhaustive polynomial-root evaluation and union-find. All dispositions match the producer selection.

The deduplicated degree-two, distinct-rational-root graph has 175 vertices, 61 connected components and 114 edges. The equality E = V - C holds (114 = 175 - 61), and the independent check records zero cycles.

Disposition counts: 114 accepted edges; 126 nonsplit or repeated-root cases; 120 signed duplicates; 60 degree skips. The input archive SHA256 is 2ebe38227ee9b2bd8cfcc460dfa0766804e650a962972a2b5054cfb74c4ef54a.

Scope: the forest claim applies only to the graph produced by those gates. Removed duplicates and known-answer label assignments are excluded; this review does not prove that they are incapable of producing scalar constraints. An additional relation set is needed for the planned distinct-edge cycle fixture, not for every conceivable geometric terminal fixture.

No inverse-recurrence predecessor was attempted and no scientific trial was run. The checked result does not establish a geometric scalar-return path. The original producer's failures, construction plan and receipts remain unchanged.

Provenance: check.py, result.json, edges.json and PLAN.md were produced by the root reviewer. This subdirectory contains presentation artifacts rendered by the Kummer agent from the existing result; rendering did not modify or rerun the independent checker. File hashes in manifest.json bind the report to those sources.